Deconstructing the Economic Growth of West Germany in the Postwar Period — How valuable were Human Capital and Immigration for the "Wirtschaftswunder" ?

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West Germany's rapid economic expansion in the aftermath of World War II (Figure 1) presents a valuable case for economic modelling because the recovery was driven, above all, by the mobilisation and enhancement of human capital. As Abelshauser (2004) argues, the country's pre-existing stock of skilled labour and technical expertise allowed it to translate institutional reforms into immediate productivity gains. Furthermore, the reconstruction period expanded Germany's robust training systems and networks, enabling workers to adapt quickly to new technologies and production methods (Berghahn, 1986). These dynamics, combined with the influx of migrant workers and the social market economy's emphasis on education and coordinated labour relations, created conditions in which human capital acted as a central engine of postwar growth (Eichengreen, 2007). Modelling this development therefore offers a way to isolate how (skilled) labour and knowledge accumulation facilitate technological adoption and long-run convergence, making West Germany a benchmark for understanding the role of human capital in the second half of the twentieth century (Temin, 2002).

In this paper, we examine whether the standard neoclassical growth model (NGM) can explain West Germany's postwar growth and provide a growth-accounting decomposition that attributes GDP per capita growth to each factor of production. In addition to a baseline specification, we explore four extensions that incorporate human capital and migration. We evaluate how alternative measurement and modelling choices affect the implied residual (TFP) and the factor contributions. Our main results show that human capital contributes positively to GDP per capita growth—especially between 1950 and 1970—and that treating human capital as a labour-augmenting factor delivers the most plausible decomposition, markedly reducing the post-1970 TFP spike relative to specifications in which human capital enters as a direct input. A wage-premium-based extension generates only minor deviations from the baseline, suggesting that efficiency

gains are more consistent with broadly shared improvements in educational attainment than with changes concentrated among a small high-skill group. Finally, a migration-based specification implies that foreign labour is quantitatively important within the accounting framework, although it does not deliver a causal estimate of immigrants' contribution.

A standard neoclassical growth model using a Cobb-Douglas production function typically specifies output as a function of physical capital and raw labour. While analytically convenient, this formulation treats all units of labour as homogeneous and therefore cannot distinguish between differences in worker skills, training, education, or health. As a result, any improvements in labour quality are absorbed into the Solow residual. This leads to an overstatement of the contribution of total factor productivity (TFP) and an understatement of factor accumulation. Introducing human capital as a separate production factor may significantly improve model accuracy for several reasons. Workers differ systematically in education, cognitive skills, and health, and these differences translate into substantial cross-country variation in productivity. By explicitly modelling the accumulation of human capital, a large portion of growth previously attributed to TFP is instead captured by changes in the quality of labour (Krueger & Lindahl, 2001).

In addition, the augmented Solow model proposed by Mankiw et al. (1992) demonstrates that, once human capital is accounted for, the model can explain roughly 80% of cross-country income variation. Economies with better-educated or more highly trained workers are more efficient at adopting new technologies, utilising complex machinery, and reorganising production processes (Hanushek & Woessmann, 2012; Mankiw et al., 1990). As shown in Figure 2, West Germany consistently outperformed the majority of European economies in terms of average years of education.

As displayed in Figure 3, West Germany saw a major influx of immigrant workers over these years. These workers were an important factor in economic growth because they expanded the labour force at a time of strong demand and facilitated structural transformation towards higher-productivity activities. Large migrant inflows helped relieve labour shortages, supported industrial expansion, and accelerated the reallocation of workers from agriculture into manufacturing and services (Braun & Kvasnicka, 2014).

The following section elaborates on the computational differences between the

models. Section two details the data sources and calibration methods. Section three presents and discusses the results. Section four concludes.

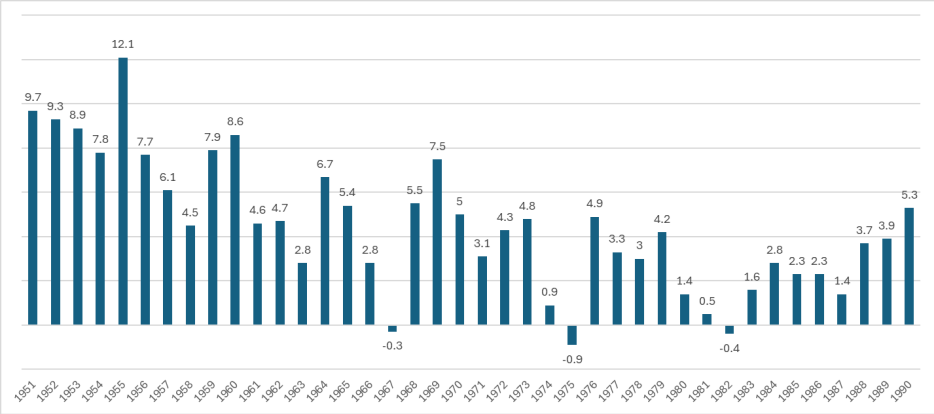

Figure 1. Economic development FRG 1950-1990. Percentage change of real GDP. Data from Statistisches Bundesamt (Destatis) (2025).

I. Models

To gain a comprehensive understanding of the role that human capital played in West Germany after 1950, we study its weight in GDP growth per unit of efficiency labour. We place our main emphasis on comparing models in order to identify the most appropriate specification. Specifically, we focus on differences in TFP to assess which model is better able to account for growth. All models are estimated using the number of workers rather than hours worked. This is because, although we observe total hours worked and the population shares of each segment, the wage-premium model would require strong additional assumptions to infer hours worked within the high-skill and low-skill groups. Given the prevalence of part-time jobs within low-skill sectors, assuming equal hours across groups would bias the results. To ensure comparability across specifications, the labour input is represented by the labour force (or relevant subgroups), and contributions to GDP growth are evaluated in terms of efficiency per worker.

We consider a sequence of production-function specifications designed to capture alternative roles of human capital and migration in the production process.

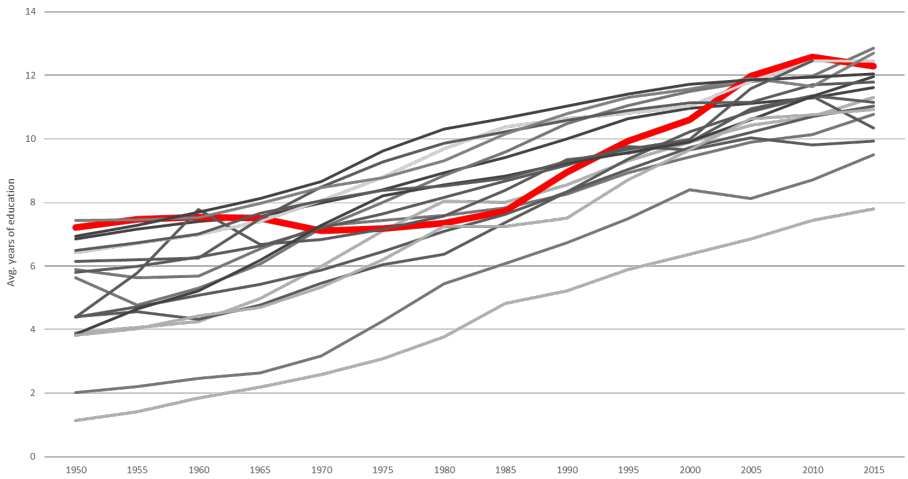

Figure 2. Average years of education in European countries from 1950-2015. Federal Republic of Germany in red. Data from Barro and Lee (2018).

Model 1 provides a baseline by assuming a standard Cobb–Douglas relationship in which output depends on total factor productivity, physical capital, and homogeneous labour. Accordingly, all workers are treated as perfect substitutes.

Model 2 relaxes this assumption by introducing human capital as a separate factor of production alongside physical capital and raw labour. In this framework, human capital is not merely an efficiency-enhancing component but a distinct input that contributes independently to output, while high- and low-skilled labour are assumed to be non-substitutable. To ensure that output does not collapse to zero in economies with low educational attainment, the human capital stock is bounded from below at one. This specification explicitly tests the validity of modelling human capital as a distinct input rather than as an augmenting factor.

Model 3 adopts a more conventional approach in which human capital augments labour. Output depends on effective labour units generated by the interaction between human capital and raw labour, implying that human capital raises the productivity of all workers proportionally. Relative to Model 2, this formulation attributes a larger share of GDP growth to labour and provides a more logically robust benchmark for assessing the contribution of human capital to aggregate output.

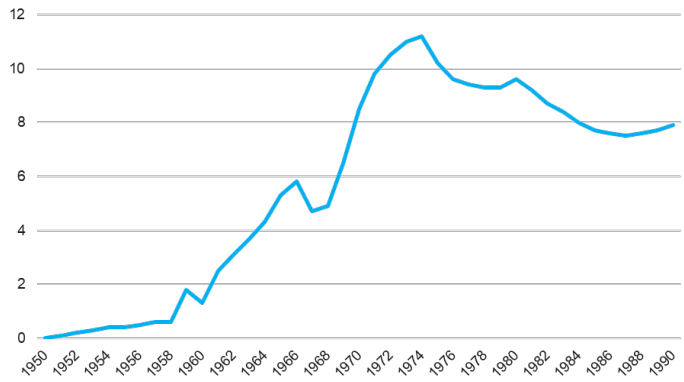

Figure 3. Foreign workers as a percentage of total employed persons over time. Data from Bundesanstalt für Arbeit.

Model 4 further refines the concept of effective labour by explicitly accounting for skill heterogeneity within the workforce. Using labour income shares, we infer relative productivity differences between high-skilled and non–high-skilled workers, measured through the wage premium defined as the ratio of labour income shares to population shares. This allows total labour input to be expressed in equivalent units of non–high-skilled labour and facilitates an assessment of whether high-skilled workers played a disproportionate role in enhancing the efficiency of the German economy.

Finally, Model 5 incorporates migration by distinguishing between native and migrant labour within the labour supply, while maintaining the assumption of perfect substitutability between the two groups. In this specification, GDP per worker is computed by dividing output by the population remaining in the counterfactual economy after migration, while physical capital is assumed to remain fixed. This approach isolates the labour-supply channel through which migration affects aggregate productivity.

Together, these models explore how human capital accumulation, skill composition, and migration shape aggregate production outcomes.

Table 1—Baseline Model and Extensions

Baseline	Human Capital	Immigration
$Y = AK^{\alpha}L^{1-\alpha}$ (1)	$Y = AK^{\alpha}(HL)^{1-\alpha}$ (2)	$Y_t = A_t K_t^{\alpha}(L_t + M_t)^{1-\alpha}$ (5)
	$Y_t = A_t K_t^{\alpha} H_t^{\beta} L_t^{1-\alpha-\beta}$ (3)	
	$Y_t = A_t K_t^{\alpha} \left(L_t^{\text{eff}}\right)^{1-\alpha}$ (4)	

II. Data and Calibration

A. *Overview and main data sources*

Given the time period on which we focus, West Germany from 1950 to 1990, data availability and reliability are central concerns. We therefore combine a small number of widely used macroeconomic datasets with historical statistical publications and archival sources to construct several key series ourselves.

As a baseline we use three main sources: the *Penn World Table* (PWT), the *Deutsche Bundesbank* long time series on national accounts, and the *Total Economy Database* (TED). For real GDP, gross investment and other aggregates we rely primarily on PWT version 5.6, which is the first PWT release to provide a consistent separation of West and East Germany back to 1950 and is therefore particularly well suited to our purposes.[1] From the Bundesbank's "Long time series on economic development in Germany" we extract detailed information on fixed capital formation by asset type and sector, as well as complementary labour-market indicators (Deutsche Bundesbank, n.d.). TED is used mainly for cross-checks and robustness (The Conference Board, n.d.).

Capital and output data from PWT 5.6 do not contain a human-capital index or a time-varying labour share. Where these are required we draw on more recent PWT releases (10.01 and 11.0), treating them as providing additional moments for unified Germany that can be mapped, with care, to West Germany. Finally, we supplement these macro sources with several more specialised datasets:

- the EU KLEMS March 2007 release for Germany, total economy (ISIC Rev. 3 code "TOT"), providing value added, labour compensation and

[1] See Feenstra et al. (2015) for a description of the PWT methodology.

labour income shares by skill group (high, medium, low) (EU KLEMS, n.d.);

- DIW/Statistisches Bundesamt tables on West German employment and wages by industry (A1/A2 series), used to approximate the skill composition of labour income before 1970 (von Knorring, 2008);
- the GESIS dataset ZA8184, *Beteiligung der Bevölkerung am Erwerbsleben in der Bundesrepublik Deutschland 1950–1997*, providing total employment and employment by broad sector (Sensch, 2004);
- historical statistics of the Bundesanstalt für Arbeit on foreign workers in the former Federal territory, which give the foreign share in employment at annual frequency from the mid-1950s onwards (Sensch, 2004).

B. *Human capital*

Human capital per worker is proxied using the PWT human-capital index. In PWT 11.0 this index is constructed as an exponential function of educational attainment by level,

$$hc_t = e^{\phi_p s_{p,t} + \phi_s s_{s,t} + \phi_t s_{t,t}},$$

where $s_{p,t}$, $s_{s,t}$ and $s_{t,t}$ denote average years of primary, secondary and tertiary schooling in the working-age population and the coefficients ϕ_p, ϕ_s, ϕ_t are constant Mincerian returns to a year of schooling at each level.[2]

PWT 5.6 does not report hc_t separately for West and East Germany, but empirical work suggests that educational attainment in the two Germanies was remarkably similar (e.g. Fraemke et al. (2025); Grätz (2021)). We therefore use the unified-Germany human-capital index from the newer PWT release as a measure of hc_t for West Germany.

C. *Constructing the productive capital stock*

PWT provides high-quality measures of real GDP, consumption and investment at constant 1985 prices, but early versions offer very limited information on the level and composition of the capital stock. In PWT 5.6 data on capital per worker begin only in 1965; separate series for residential and infrastructure capital start

[2] See the PWT 11.0 documentation for details on the construction of hc_t and the underlying Barro–Lee schooling data.(Barro & Lee, 2013)

even later. To obtain a productive capital stock suitable for growth-accounting and TFP calculations over 1950–1990 we combine Bundesbank, PWT and TED information with our own reconstruction for the pre-1961 period.

The Bundesbank's long-term national accounts report productive (non-residential, non-infrastructure) capital for West Germany beginning in 1961. Using these data together with PWT and TED we construct a benchmark level of productive capital K_{1961}^{prod} (in 1985 prices) and then extend the series both backward and forward using a perpetual-inventory equation with geometric depreciation,

$$K_{t+1}^{prod} = (1-\delta)K_t^{prod} + I_t^{prod}.$$

Following the literature we fix the depreciation rate at $\delta = 0.06$ for machinery and equipment. For completeness, residential and infrastructure capital are treated with lower depreciation rates (0.02), but these play a role only in the internal feasibility checks described below.

A key difficulty is that reliable information on the composition of gross investment by asset type is scarce in the 1950s. PWT investment is aggregate and cannot be cleanly split into productive, residential and infrastructure components. Using aggregate investment directly in the PIM would therefore overstate the productive capital stock that enters the production function. At the same time, the standard optimisation approach that infers the initial capital stock by imposing a steady-state capital–output ratio is ill-suited to the highly non-stationary conditions of post-war West Germany. We therefore consider two alternative reconstruction strategies, both anchored on K_{1961}^{prod} but embodying different assumptions about the state of the capital stock in 1950.

Model I (*K_Model*): reconstruction hypothesis

The first approach assumes that productive capital in West Germany was severely damaged by the war and that a large share of investment in the 1950s was devoted to rebuilding it. Guided by Abelshauser (2004), we posit that a constant share θ of aggregate gross investment goes into productive capital,

$$I_t^{prod} = \theta I_t^{total},$$

with θ calibrated to 0.46.[3] Starting from the observed 1961 stock K^{prod}_{1961} and applying the PIM backwards, we obtain a time series $\{K^{prod}_t\}^{1961}_{1950}$ consistent with the assumed depreciation rate and investment share. For $t \geq 1961$ we evolve the capital stock forwards using the same equation.

This method delivers an initial productive capital–output ratio in 1950 of $K^{prod}_{1950}/Y_{1950} \approx 0.28$, in line with Dumke (1990)'s estimates that place it between 0.25 and 0.30 of GDP. In our subsequent growth-accounting exercises this specification implies rapid post-war capital deepening and relatively moderate TFP growth.

Model II (*K_Optim*): not-so-damaged hypothesis

The second approach embodies the opposite view from the "German economic miracle" literature, namely that capital was not heavily destroyed and that the post-war boom primarily reflected productivity gains (Vonyó, 2020). In this case we fix the initial productive capital–output ratio to a much higher level,

$$\frac{K^{prod}_{1950}}{Y_{1950}} = 1.75,$$

and treat the share of investment flowing into productive capital as an unknown to be determined endogenously. Given K^{prod}_{1950}, $\delta = 0.06$ and the 1961 benchmark K^{prod}_{1961}, we solve for a sequence of investment shares that makes the PIM path hit the 1961 target. The implied average investment share into productive capital is about 24% of total gross investment.

Model comparison

The two reconstructed capital series, denoted K^{M1}_t and K^{M2}_t, differ markedly. Model I yields a low initial capital stock that rises smoothly as investment accelerates; Model II implies an already high stock in 1950 and much smaller net capital accumulation thereafter. To assess which reconstruction is more plausible we subject both series to two validation exercises: (i) feasibility of investment decomposition and (ii) qualitative implications for aggregate growth and TFP.

First, we allocate total investment across three asset types—productive, residential and infrastructure capital—under the constraint that investment shares

[3] Abelshauser (2004) provides detailed evidence on the sectoral and asset composition of post-war investment in West Germany.

must sum to one in each year. Starting from plausible initial values for residential and infrastructure capital (e.g. $K^{res}_{1950}/Y_{1950} = 0.5$, $K^{infra}_{1950}/Y_{1950} = 0.07$; Piketty and Zucman (2014)), we ask whether the implied investment shares are economically reasonable. Under Model II, even extreme assumptions (setting residential and infrastructure capital to zero in 1950) fail to reconcile the high initial K^{prod}_{1950} with observed investment flows: the model can explain only about 72% of total investment when non-negative shares are imposed, and the required patterns of investment into housing and infrastructure are implausibly low. Under Model I, by contrast, the reconstructed series can account for roughly 85–93% of total investment with investment shares in residential and infrastructure capital (around 30% and 13–14% respectively in the early 1950s) that are consistent with Bundesbank reports and independent estimates.

Second, we embed each capital series in a standard Solow-style growth-accounting decomposition of GDP per worker, using the same labour and TFP calibration described below. As shown in Figures 4, 5 and 6, Model I produces a smooth and economically intuitive evolution of the capital stock, a moderate TFP residual and a growth decomposition in which capital deepening plays an important, but not dominant, role. Model II, in contrast, implies unrealistically low contributions from capital and extremely high TFP growth, effectively attributing the post-war recovery almost entirely to productivity shocks. Together, these findings lead us to adopt Model I (*K_Model*) as our baseline measure of productive capital.

Using the Model I series for the total capital stock, we also compute the overall capital–output ratio in 1950 and obtain $K_{1950}/Y_{1950} \approx 0.85$, which is consistent with earlier work suggesting that the aggregate capital–output ratio in post-war West Germany was below one (Maddison (1991); Goldsmith (1985)). In light of these results we use *K_Model* for all subsequent calibrations.

D. *Factor-income decomposition: capital vs. skill groups*

We next construct time-varying factor-income shares for capital, high-skill labour and other labour. Denote by A_t the capital share, by B_t the share of income accruing to high-skill workers, and by $1 - A_t - B_t$ the share accruing to medium- and low-skill workers.

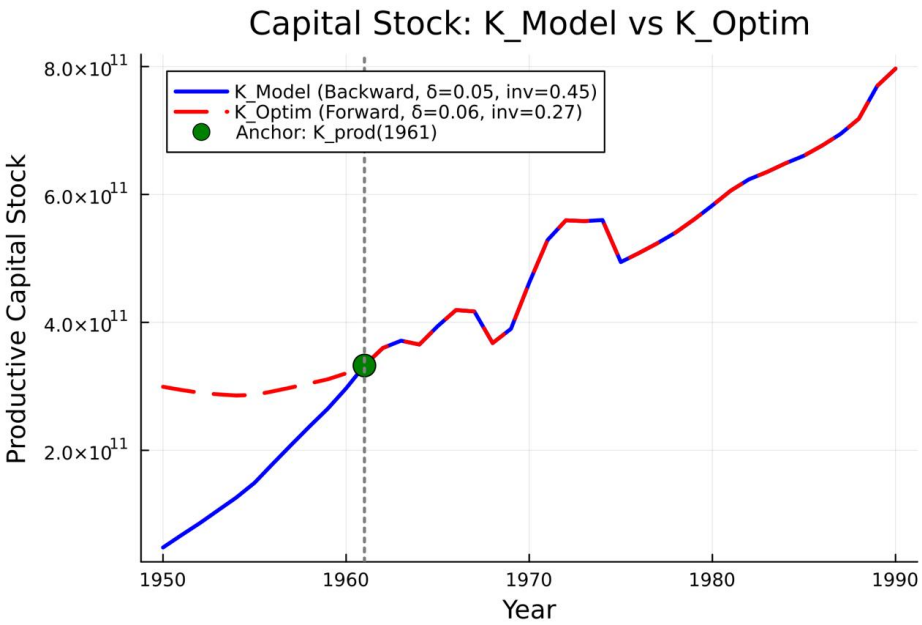

FIGURE 4. COMPARISON OF RECONSTRUCTED CAPITAL SERIES, MODEL I (*K_Model*) VS. MODEL II (*K_Optim*).

LABOUR AND CAPITAL SHARES

For 1950–1969 we take the labour share in GDP, $labsh_t$, from PWT and define

$$A_t = 1 - labsh_t, \qquad t = 1950, \ldots, 1969.$$

Over this period the labour share is close to two-thirds, consistent with the standard Kaldor fact.

From 1970 onwards we use EU KLEMS, which provides both value added, VA_t, and labour compensation, LAB_t, for the total economy. We construct the labour share as

$$labour_share_t = \frac{LAB_t}{VA_t},$$

and define

$$A_t = 1 - labour_share_t, \qquad t = 1970, \ldots, 1990.$$

The splice between PWT and EU KLEMS preserves continuity in the aggregate

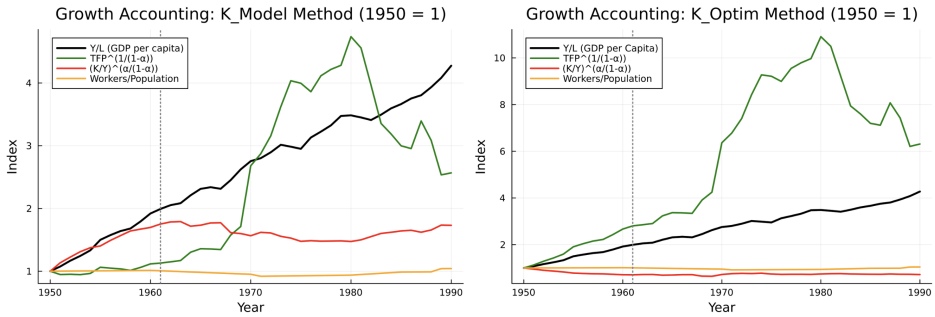

FIGURE 5. COMPARISON OF IMPLIED GROWTH ACCOUNTING, MODEL I (*K_Model*) VS. MODEL II (*K_Optim*)

labour share while allowing for modest time variation driven by the KLEMS data.

HIGH-SKILL INCOME SHARE B_t

The high-skill income share is obtained in two steps, reflecting a change in data availability around 1970.

PRE-1970: HIGH-WAGE INDUSTRIES AS A PROXY.

Before 1970, no direct breakdown of labour compensation by skill exists. We therefore use the wage structure across industries as a proxy for the skill distribution. Based on the A1/A2 tables in von Knorring (2008) for West Germany we compute, for each industry j and year t,

$$W_{j,t} = L_{j,t}\, w_{j,t},$$

the wage bill of industry j. We rank industries by their average wage over 1950–1970,

$$\bar{w}_j = \frac{1}{21} \sum_{t=1950}^{1970} w_{j,t},$$

and select the four industries with the highest $\bar{w}_j$ as a proxy for the high-skill segment of the industrial workforce: Mineralölverarbeitung, Zuckerindustrie, Eisenschaffende Industrie and Erdöl-/Erdgasgewinnung. Let $\mathcal{H}$ denote this set of "top-

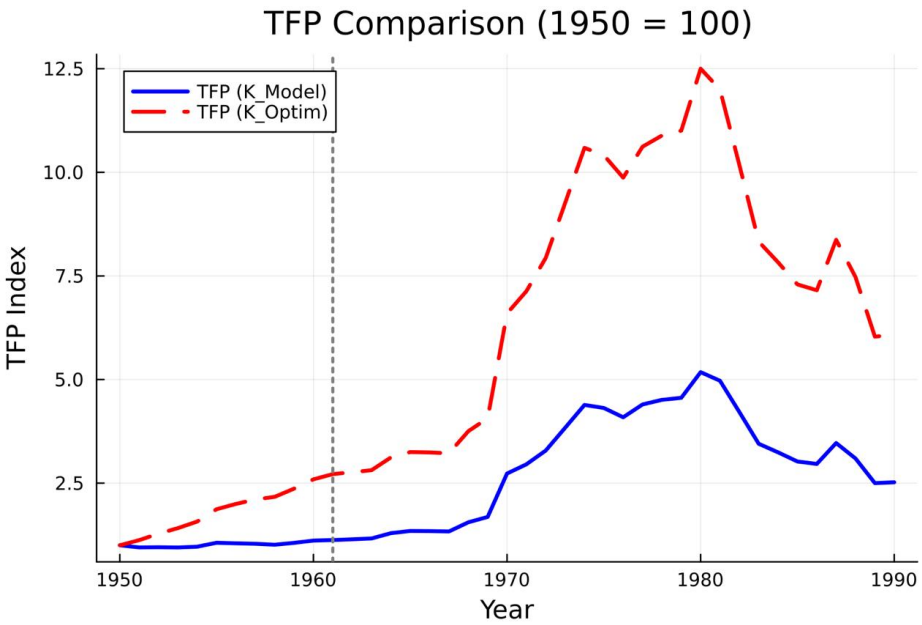

FIGURE 6. COMPARISON OF TFP, MODEL I (*K_Model*) VS. MODEL II (*K_Optim*).

4" industries. The wage-bill share of these high-wage industries is

$$\beta_t^{(w)} = \frac{\sum_{j\in\mathcal{H}} W_{j,t}}{\sum_j W_{j,t}}, \qquad t = 1950, \ldots, 1970.$$

The size of $\mathcal{H}$ is chosen so that $\beta_{1970}^{(w)}$ is close to the high-skill labour-income share reported by EU KLEMS (about 8% of total labour compensation), ensuring continuity at the splice point. The share of GDP accruing to high-skill labour before 1970 is then

$$B_t = labsh_t\, \beta_t^{(w)}, \qquad t = 1950, \ldots, 1969.$$

POST-1970: EU KLEMS SKILL DECOMPOSITION.

From 1970 onwards, EU KLEMS provides the share of labour compensation going to high-skill workers, $LABHS_t$ (in percent). Combining this with the labour

share yields

$$B_t = labour_share_t \times \frac{LABHS_t}{100}, \qquad t = 1970, \ldots, 1990.$$

For all years we define the residual labour-income share as

$$1 - A_t - B_t,$$

which can be interpreted as the income share of medium- and low-skill labour (plus any remaining measurement residual).

E. High-skill employment (headcount)

For descriptive purposes we also construct a headcount series for high-skill workers.

1950–1970: TOP-4 INDUSTRY EMPLOYMENT.

Using A1 and GESIS ZA8184 we define high-skill employment as the number of workers employed in the four high-wage industries,

$$N_t^H = \sum_{j \in \mathcal{H}} L_{j,t},$$

and total employment in the economy as N_t^{tot} ("Alle Wirtschaftsbereiche — Undifferenziert"). The raw high-skill share is then

$$share_t^{H,pre} = \frac{N_t^H}{N_t^{tot}}, \qquad t = 1950, \ldots, 1970.$$

Because N_t^H covers only a subset of industries (predominantly manufacturing), this raw share understates the true share of high-skill workers in the economy. To align levels with the EU KLEMS employment-based skill shares, we multiply N_t^H by a constant factor c such that the implied high-skill share in 1970 matches the EU KLEMS high-skill employment share, $share_{1970}^{H,KLEMS}$:

$$c = \frac{share_{1970}^{H,KLEMS}}{share_{1970}^{H,pre}}, \qquad \tilde{N}_t^H = cN_t^H, \qquad \widetilde{share}_t^{\,H} = \frac{\tilde{N}_t^H}{N_t^{tot}}.$$

1970 ONWARDS: EU KLEMS HEADCOUNT PROXY.

For 1970 and later we compute a high-skill headcount directly from EU KLEMS as

$$N_t^{H,KLEMS} = N_t^{tot,KLEMS} \times \frac{HIGHSKILL_SHARE_t}{100},$$

where $N_t^{tot,KLEMS}$ is total employment and $HIGHSKILL_SHARE_t$ is the high-skill employment share. The scaled pre-1970 series $\tilde{N}_t^H$ is then spliced to $N_t^{H,KLEMS}$ at 1970, yielding a smooth high-skill headcount series from 1950 to 2004.

F. Native and migrant labour input

To allow for different elasticities of native and migrant labour we split total hours worked into native hours L_t and migrant hours M_t. Total hours are obtained from PWT as

$$H_t = emp_t \times avh_t,$$

where emp_t is the number of persons engaged (millions) and avh_t is average annual hours worked per person.

From the Bundesanstalt für Arbeit statistics we take the series "foreign workers as a percentage of total employment" in the former Federal territory, denoted $foreign_share_t$ (in percent). This series is available annually from 1954 onwards; for 1950–1953 we linearly interpolate backwards from the first observed value, imposing $foreign_share_{1950} = 0$. Defining the foreign employment share in fractional terms,

$$s_t^F = \frac{foreign_share_t}{100},$$

and assuming that foreign and native workers supply the same average number of hours, we split total hours as

$$M_t = s_t^F H_t, \qquad L_t = (1 - s_t^F) H_t.$$

These series enter the Cobb–Douglas specification with migrant labour,

$$Y_t = A_t K_t^{\alpha} (L_t + M_t)^{1-\alpha},$$

Taken together, the procedures described above yield a unified dataset for West Germany, 1950–1990, containing consistent series for the productive capital stock,

human capital, native and migrant labour inputs, and a factor-income decomposition into capital, high-skill labour and other labour. These are the objects used in the quantitative analysis that follows.

III. Results

The difference displayed in the interaction of the five models with the gathered data is a first significant result of the study. It points out that the methods used to capture the same variable (Model 5 excluded) play a crucial role in determining the explicative value of the extension it-self. Comparing each model with the baseline of Model 1 suggests that the implementation of human capital has a sound informative significance over the German economic miracle growth decomposition. The biggest finding is outlined by the positive contribution to GDP per capita growth that human capital has between 1950 and 1970, in all his specifications. This supports our initial hypothesis and strengthens the idea that the highly damaged German physical capital was counterbalanced by a considerable amount of human capital stock. This, we believe, has enabled the economy to undergo a growth miracle and exploit the high levels of foreign and local investment received. Dissecting the results more narrowly, we found in Model 2 an interesting and almost unprecedented way to model human capital which nevertheless needs to be improved, in Model 3 our preferred specification and in Model 4 an expectable result that enhances the whole analysis. A different analysis must be carried over Model 5. Compared with the baseline, Model 2 performs fairly well over the years 1950 to 1970, but the post-1970 spike in TFP may indicate two potential problems. Either the model fails to capture major structural changes in the economy—so that human capital no longer accounts for the implied decline in physical capital—or the specification itself is misspecified. Mankiw et al. (1992) estimate the return to human capital and find a value of β close to 0.33. Interpreting the return to high-skill labour as the return to human capital is economically defensible, but it yields a value roughly an order of magnitude smaller than Mankiw et al.'s estimate, implying an economy in which human capital plays a more limited role in growth. More generally, the difficulty of measuring human capital (relative to labour and physical capital) and identifying its associated return casts doubt on treating it as a direct input in production. By contrast, endogenous-growth approaches that view human capital as a source of efficiency improvements and as a factor shaping labour–leisure–education choices

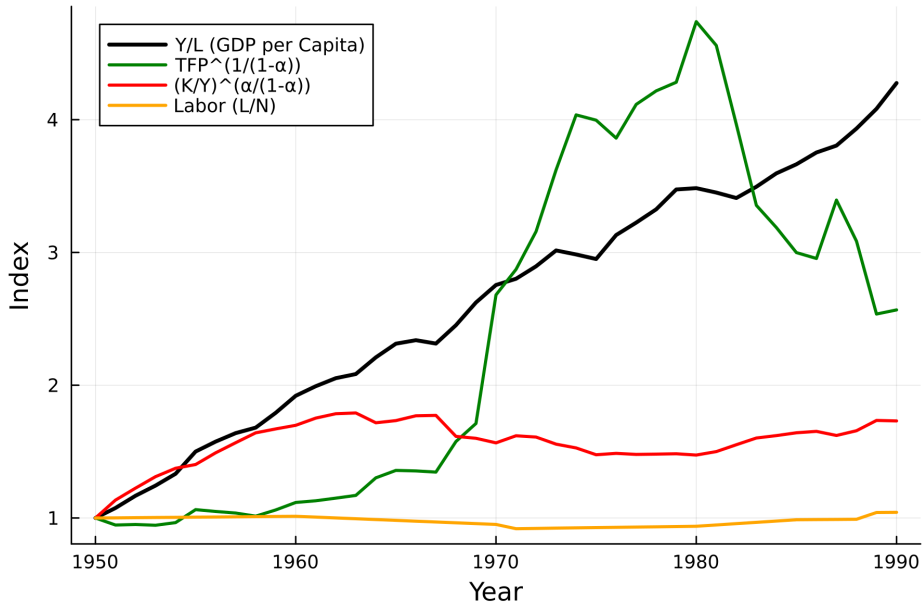

Figure 7. Model 1: Baseline Growth Decomposition

(**Lucas1988**) have pushed the literature toward our third model, in which human capital enters as a labour-augmenting factor. Model 3 yields the lowest TFP. Over the full sample period, it performs better than our baseline and aligns closely with the existing literature. The model suggests that the rising contribution of labour to GDP growth between 1950 and 1990 is largely driven by expanded access to education—especially growth in tertiary educational attainment. This pattern is not captured by Model 1, which—because it is unable to assess the increase in efficiency per worker—implies a roughly constant contribution of labour to GDP growth. The gap in implied TFP across specifications therefore provides additional evidence of an unaccounted-for labour-augmenting effect of human capital.

The smaller post-1970 TFP spike offers a further insight. When human capital is treated as a direct input, it does not fully absorb the decline in the physical capital stock around 1967; by contrast, the labour-augmenting specification performs better, even relative to the standard model. Put differently, improvements in the productivity of (on average) more highly educated workers appear better able to offset lower capital intensity associated with reduced investment flows—something

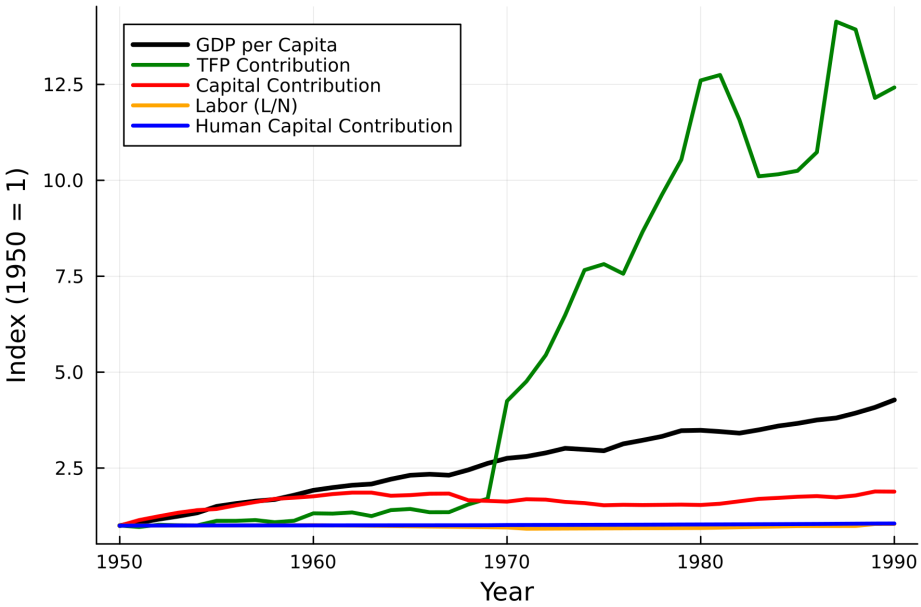

Figure 8. Model 2: Human Capital as Input Growth Decomposition

the alternative specifications fail to capture. This mechanism provides our most direct explanation for the core research question: higher educational attainment raised efficiency and thereby counterbalanced war-damaged capital and subsequent contractions in capital over the course of the economic miracle. Model 4 performs relatively poorly, in the sense that it differs only marginally from the baseline specification in Model 1. This outcome was expected and adds an important piece to the puzzle. We examine whether high-skill workers account for a substantial share of the increase in German efficiency by constructing a "wage premium" measure defined as the high-skill wage-bill share divided by the high-skill population share. The results suggest this is not the case: Model 4 shows only a minor deviation from Model 1 over the 40-year period.

It is therefore plausible that the rise in labour efficiency documented in Model 3 reflects broadly shared improvements in educational attainment across the population, rather than gains concentrated among a small, highly educated elite driving aggregate growth. Cross-country comparative studies of educational systems, such as Müller and Karle (1993) and Lutz (1976), similarly contrast West Ger-

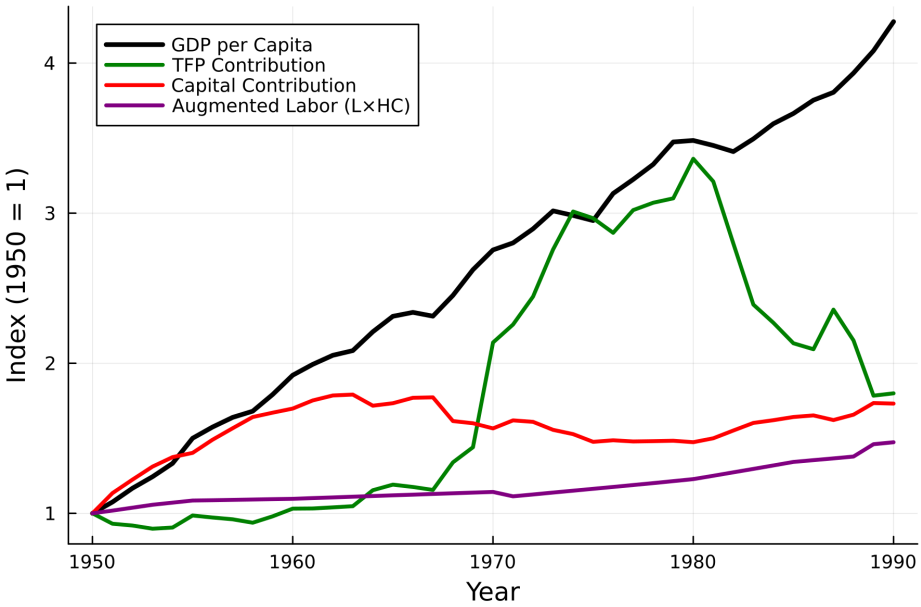

Figure 9. Model 3: Human Capital as Labour Augmenting Factor Growth Decomposition

many's relatively widespread educational attainment with the French, American, and British systems, which they describe as more polarized—combining highly specialized managerial groups with a larger low-skilled workforce.

Counterfactually, in an economy closer to those cases, we would expect a different pattern: the labour-augmenting human-capital channel captured in Model 3 would account for a smaller share of GDP per capita growth, while a wage-premium mechanism of the type summarized by Model 4 would matter more. Model 4 therefore acts as a robustness check for our preferred human-capital specification and is consistent with institutional accounts emphasizing broadly distributed skills and relatively compressed returns to education in West Germany (Hall & Soskice, 2001). Model 5 behaves in line with our expectations. Mechanically, a smaller contribution of labour to GDP per capita growth and a larger residual component (TFP) follow when a smaller population is assumed to operate the same stock of productive capital. However, drawing strong inferences about the role of immigrants in West Germany's economic miracle remains difficult. To identify the impact of foreign workers on GDP per capita growth,

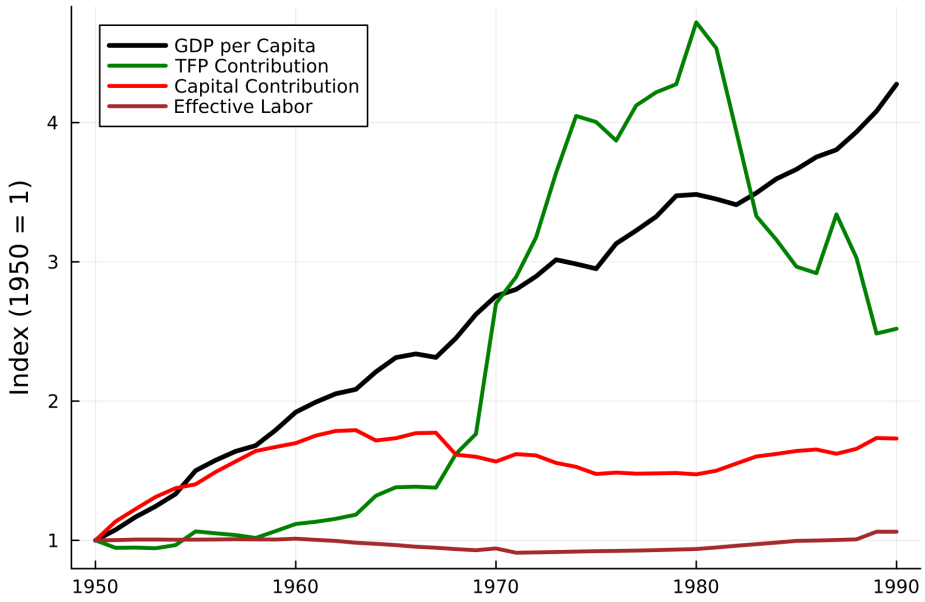

Figure 10. Model 4: Wage Premium Model

we would need a credible counterfactual describing what would have occurred in their absence—something our model cannot provide for two reasons.

First, the assumption that native and foreign workers supply perfectly substitutable types of labour requires substantially more justification (De New & Zimmermann, 1994). Second, if—absent immigrant workers—the remaining native labour force had to supply the same aggregate amount of labour, their measured efficiency would likely have increased. In that case, the observed difference in TFP between the two scenarios would have been at least partially absorbed by changes in the labour-income share attributed to the remaining native workforce.

At the same time, this basic economic intuition is complicated by empirical evidence suggesting that reliance on foreign workers enabled natives to specialize further and move into higher-skilled and better-paid occupations (Herbert, 1990). Dohse (1996) similarly argues that foreign labour supply increased labour-market elasticity, often helping to protect native workers' positions during economic downturns.

Because we do not have reliable data on the returns to immigrant labour vis-

à-vis native labour, nor on nativity-specific wage-bill and hours-worked shares, we refrain from a causal interpretation. Nevertheless, Model 5 delivers a clear conditional implication: under the maintained assumption that capital and efficiency are held constant, the observed growth path of GDP per capita is difficult to reconcile with an economy in which foreign labour is removed without replacement. In other words, immigrant labour appears to be a quantitatively important margin in the growth-accounting decomposition, even though its causal contribution cannot be pinned down without modelling imperfect substitution and the endogenous responses of natives' labour supply and occupational choice.

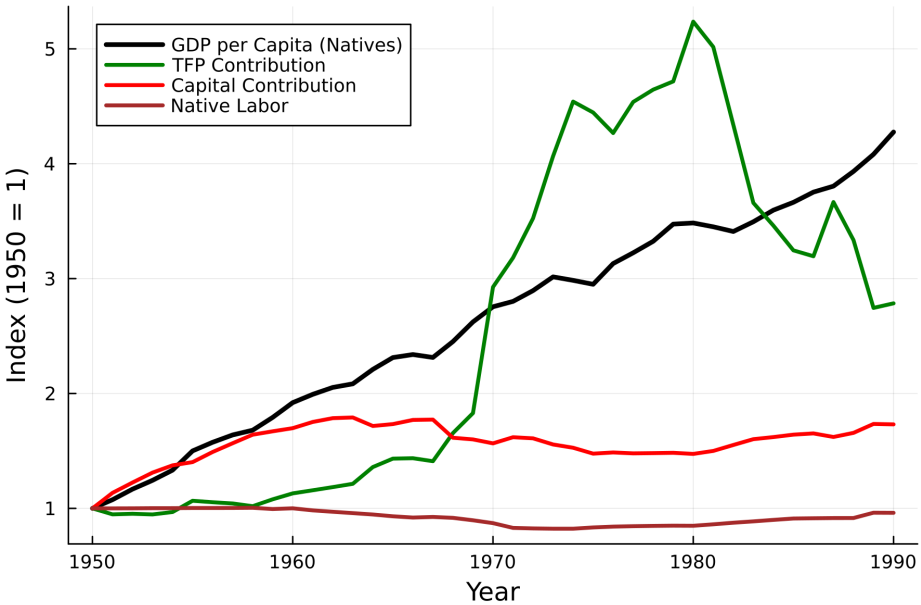

Figure 11. Model 5: Natives Labour Only Growth Decomposition

IV. Conclusion

This paper set out to reassess the sources of West Germany's postwar growth between 1950 and 1990 by confronting the standard Cobb–Douglas growth-accounting framework with a sequence of extensions designed to better reflect the historical narrative of the "Wirtschaftswunder". The central research question was whether

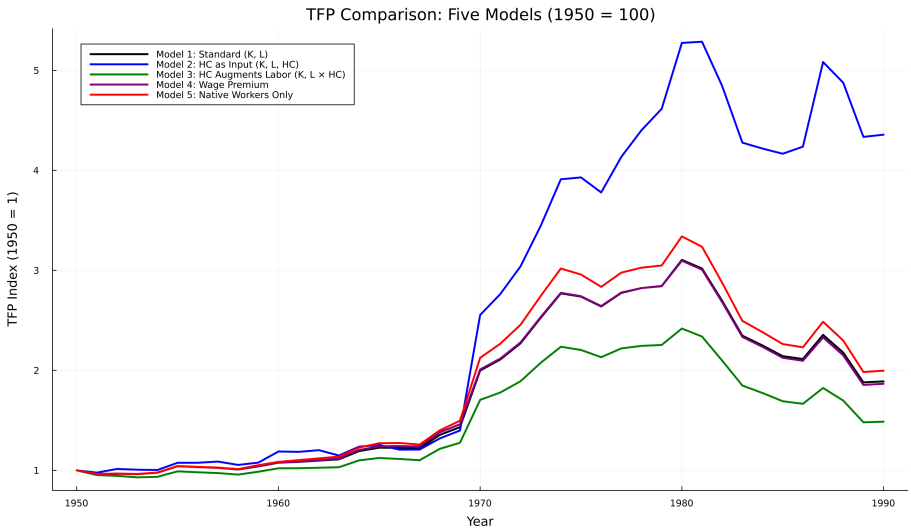

FIGURE 12. TFP COMPARISON ACCROS MODELS

the rapid recovery can be explained primarily by factor accumulation—particularly increases in human capital and labour supply—or whether it must be largely attributed to residual total factor productivity.

Three conclusions emerge from the comparison of the five models. First, relative to the baseline specification, incorporating human capital improves the informational content of the decomposition. Across all human-capital specifications, human capital contributes positively to GDP per capita growth in the key recovery phase between 1950 and 1970. This supports the paper's central hypothesis: the legacy of war-damaged physical capital can be reconciled with rapid output growth only if labour quality rose meaningfully in the early decades of reconstruction. In this sense, human capital helped counterbalance limited productive capital and facilitated the economy's capacity to absorb both domestic and foreign investment.

Second, the model comparison indicates that how human capital is introduced matters at least as much as whether it is included at all. Treating human capital as a direct factor input performs reasonably well up to 1970 but generates an implausible post-1970 TFP spike, suggesting that the specification either misses key structural changes or misallocates shifts in capital intensity and labour efficiency

to TFP. By contrast, the labour-augmenting human-capital specification yields the lowest and most stable TFP residual and produces a more coherent narrative of West German growth. Under this preferred specification, rising educational attainment across the labour force—especially at higher levels—raises efficiency per worker and thereby better offsets periods of declining capital intensity. This provides the most direct answer to the core research question: improvements in educational attainment operated primarily through labour-augmenting efficiency rather than through a large measured "return to human capital" as a separate input.

Third, the additional extensions aid this interpretation. The wage-premium model differs only marginally from the baseline, implying that West Germany's improvement in labour efficiency was unlikely to have been driven mainly by a small, highly educated elite. Instead, the evidence is consistent with broadly shared increases in educational attainment and relatively compressed returns to education. The migration exercise produces a clear accounting implication—removing foreign labour mechanically lowers the labour contribution and raises the residual—but it does not permit a causal interpretation. Quantifying the contribution of immigrants to growth would require modelling imperfect substitution between native and foreign workers, and incorporating nativity-specific wages, hours, and endogenous responses of natives' labour supply and occupational choice.

Taken together, the results suggest a coherent interpretation of the West German economic miracle. Physical capital deepening mattered, but a growth-accounting framework that ignores human capital places too much weight on TFP. Once human capital is introduced in a way consistent with its role as a driver of overall worker efficiency, the postwar growth experience is more plausibly explained as a combination of capital accumulation and these broad improvements in labour quality, with immigrant labour playing an important quantitative margin but remaining difficult to evaluate causally with the available data.

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Appendix

https://github.com/williammcfrln/MacroEconomics-Project.git

This link contains the data, Julia code, and instructions needed to replicate our paper.